

Nikhil D'Souza

dsouza13@purdue.edu | (805) 490-5055 | nikhiljay.com | github.com/nikhiljay | Eagle Scout

Education

Purdue University – West Lafayette, IN

May 2022

- B.S. in Computer Science & B.S. in Data Science (GPA: 3.85)
- Coursework: Data Mining & Machine Learning, Analysis of Algorithms, Intro to AI, Data Structures & Algorithms, Applied Regression Analysis, Systems Programming, Statistical Theory, Probability, Discrete Math, Linear Algebra, Multivariable Calculus

Coursera – deeplearning.ai

2020

- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, & Optimization

Experience

Vincere Health – Harvard Innovation Lab

Oct 2019 - Aug 2020

Software Engineer & Data Scientist

- Built a platform that uses financial incentives, behavioral nudges, and evidence-based interventions to help quit smoking and be healthy.
- Developed key features such as patient facial recognition, chat messaging using sockets, and incentive engine.
- Spearheaded data science capability providing actionable insights, user performance, and patient rewards.

Taco Bell Corp. – Irvine, CA

May 2019 - Aug 2019

Software Engineering Intern

- Built real-time dashboards using ELK stack to improve digital oversight and business strategy across sales and operations.
- Worked in an innovation lab on a next-gen kitchen display system currently rolling out across 10k restaurants in the US.
- Presented a potential next-gen business operating model using modern technologies at a company-wide event (recognized by CIO).

Viasat Inc. – Carlsbad, CA

Oct 2018 - Apr 2019

Data Science Intern

- Built an anomaly detection framework that utilizes unsupervised machine learning to detect novel patterns in time series data.
- Used the Generalized ESD algorithm with seasonal decomposition to detect faulty equipment, congestion, and usage anomalies.
- Presented to leadership about proactively managing service interruptions by predicting satellite and hub downtimes.

Saama Technologies Inc. – Campbell, CA

Jun 2018 - Aug 2018

AI Intern

- Used machine learning techniques to identify packaging anomalies on a conveyor belt for a leading biotech company.
- Developed CNN with 97% validation accuracy and leveraged image augmentation to expand the dataset.
- Resulted in the global rollout of AI/ML capability in manufacturing facilities and optimized delivery of the product to patients.

Projects & Awards

Feverbase.org – COVID clinical trial data platform

Jun 2020

- Built a platform to connect researchers, patients, and the public to relevant information for drug discovery and trial participation.

Harvard President's Innovation Challenge - Grand Prize Winner

May 2020

- Our startup received \$75k for using real-time incentives and behavioral nudges to motivate people to live healthier lives.

Harris Corporation Scholarship – Purdue CS Awards 2019

Apr 2019

- Awarded for excellent academic performance.

Calculating Fatigue of Rugby Athletes – ASA DataFest 2019 (1st Place Overall Winner)

Mar 2019

- Determined real-time fatigue levels of rugby players by identifying impacts, measuring sprint distance, and analyzing health metrics.

GluClose – PennApps XVIII (2nd Place Overall Winner)

Sep 2018

- Built a hack that uses saliva and bacteria in mud to quickly tell if blood glucose concentration is abnormally high.

Apple Worldwide Developer Conference Scholarship Winner

Jun 2015

- One of 350 students selected globally to attend the conference based on the submitted app and proficiency in iOS and Swift.

Skills

Languages/Tools: Python, R, SQL, Tensorflow, Pytorch, Flask, React, Typescript, JS ES6, Node.js, Swift, Java, C, Unix, Git, Docker, Redis

AWS Services: EC2, S3, Lambda, API Gateway, Sagemaker, Elasticsearch, CloudWatch, Rekognition, Comprehend, SNS, Cognito